

ECEN-743: Homework 1

Due on February 8, 2026 at 6:00 PM

Instructor: Dr. Dileep Kalathil

Brody Jordan
431005357

Problem 1

Let $x, y \in \mathbb{R}^n$. The triangle inequality states that $\|x + y\| \leq \|x\| + \|y\|$. Use this to show that $\|x - y\| \geq \|x\| - \|y\|$.

Solution

Starting with

$$\|x + y\| \leq \|x\| + \|y\|$$

and setting $x = (x - y) + y$, we get

$$\|(x - y) + y\| \leq \|x - y\| + \|y\|$$

and rearranging gives

$$\|x - y\| \geq \|(x - y) + y\| - \|y\| = \|x\| - \|y\|$$

Problem 2

Consider an MDP with discount factor $\gamma \in (0, 1)$. Show that

$$\sup_{\pi} \|V_{\pi}\|_{\infty} \leq \frac{\max_{s,a} |r(s, a)|}{(1 - \gamma)}.$$

Solution

Value function V_{π} is defined as

$$V_{\pi}(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s, \pi \right]$$

and thus

$$|V_{\pi}(s)| \leq \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t |r(s_t, a_t)| \mid s_0 = s, \pi \right] \leq \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \max_{s,a} |r(s, a)| \mid s_0 = s, \pi \right] = \frac{\max_{s,a} |r(s, a)|}{(1 - \gamma)}$$

which implies that

$$\sup_{\pi} \|V_{\pi}\|_{\infty} = \sup_{\pi} \max_s |V_{\pi}(s)| \leq \frac{\max_{s,a} |r(s, a)|}{(1 - \gamma)}.$$

Problem 3

Show that the Bellman operator T is a monotone operator, i.e, for any $V_1, V_2 \in \mathbb{R}^{|S|}$ with $V_1 \geq V_2$ (elementwise), $TV_1 \geq TV_2$.

Solution

The Bellman operator is defined as

$$(TV)(s) = \max_a (r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V(s')])$$

and since $V_1 \geq V_2$

$$\max_a (r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V_1(s')]) \geq \max_a (r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V_2(s')])$$

which is simply $TV_1 \geq TV_2$.

Problem 4

Consider the function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $f(u) = Au$, where $A \in \mathbb{R}^n \times \mathbb{R}^n$. Assume that the row sums of A is strictly less than 1, i.e., $\sum_j |a_{ij}| \leq \alpha < 1$. Show that $f(\cdot)$ is a contraction mapping with respect to $\|\cdot\|_\infty$.

Solution

With respect to $\|\cdot\|_\infty$, we have

$$\|f(u) - f(v)\|_\infty = \|A(u - v)\|_\infty$$

which must be less than or equal to $\alpha \|u - v\|_\infty$ to be a contraction mapping. Rewriting this gives

$$\max_i |(A(u - v))_i| = \max_i \left| \sum_{j=1}^n a_{ij}(u - v)_j \right| \leq \max_i \sum_{j=1}^n |a_{ij}| \cdot |(u - v)_j|$$

which we can use to write

$$\|f(u) - f(v)\|_\infty \leq \alpha \|u - v\|_\infty$$

and thus, $f(\cdot)$ is a contraction mapping with respect to $\|\cdot\|_\infty$.

Problem 5

Let $\mathcal{U}$ be a given set, and $g_1 : \mathcal{U} \rightarrow \mathbb{R}$ and $g_2 : \mathcal{U} \rightarrow \mathbb{R}$ be two real-valued functions on $\mathcal{U}$. Also assume that both functions are bounded. Show that

$$|\max_u g_1(u) - \max_u g_2(u)| \leq \max_u |g_1(u) - g_2(u)|$$

Solution

Starting with

$$g_1(u) = g_2(u) + (g_1(u) - g_2(u))$$

and taking the maximum and abs on the RHS

$$g_1(u) \leq g_2(u) + \max_u |g_1(u) - g_2(u)|$$

then rearranging and taking the maximum on the LHS gives

$$|\max_u g_1(u) - \max_u g_2(u)| \leq \max_u |g_1(u) - g_2(u)|$$

Problem 6

Consider the value iteration algorithm $V_{k+1} = TV_k$, with an arbitrary V_0 , where T is the Bellman operator.

(a) Show that, for $n > m$

$$\|V_m - V_n\|_\infty \leq \frac{\gamma^m}{(1-\gamma)} \|V_0 - V_1\|_\infty$$

(b) Let V^* be the optimal value function. Show that

$$\|V_m - V^*\|_\infty \leq \frac{\gamma^m}{(1-\gamma)} \|V_0 - V_1\|_\infty$$

(c) Show that

$$\|V_m - V^*\|_\infty \leq \frac{\gamma}{(1-\gamma)} \|V_{m-1} - V_m\|_\infty$$

Solution

Part (a): When $n > m$

$$\|V_m - V_n\|_\infty = \left\| \sum_{k=m}^{n-1} (V_k - V_{k+1}) \right\|_\infty \leq \sum_{k=m}^{n-1} \|V_k - V_{k+1}\|_\infty$$

which through factorization and evaluation of the sum we can write

$$\|V_m - V_n\|_\infty \leq \frac{\gamma^m}{(1-\gamma)} \|V_0 - V_1\|_\infty$$

Part (b): Using the properties of the Bellman operator, we have

$$\lim_{n \rightarrow \infty} \|V_m - V_n\|_\infty = \|V_m - V^*\|_\infty \leq \frac{\gamma^m}{(1-\gamma)} \|V_0 - V_1\|_\infty$$

Part (c): Since $V_{k+1} = TV_k$,

$$\|V_{m+1} - V^*\|_\infty = \|TV_m - TV^*\|_\infty \leq \gamma \|V_m - V^*\|_\infty$$

and thus

$$\|V_m - V^*\|_\infty \leq \frac{\gamma}{(1-\gamma)} \|V_{m-1} - V_m\|_\infty$$

Problem 7

Let $\bar{Q}$ be such that $\|\bar{Q} - Q^*\|_\infty \leq \epsilon$, where Q^* is the optimal Q -value function. Let $\bar{\pi}$ be the greedy policy with respect to $\bar{Q}$, i.e., $\bar{\pi}(s) = \arg \max_a \bar{Q}(s, a)$. Show that

$$\|V^* - V_{\bar{\pi}}\|_\infty \leq \frac{2\epsilon}{(1-\gamma)}$$

Solution

In this case we have

$$V^*(s) - V_{\bar{\pi}}(s) = Q^*(s, \pi^*(s)) - Q^*(s, \bar{\pi}(s))$$

and since $\bar{\pi}(s) = \arg \max_a \bar{Q}(s, a)$, we have

$$Q^*(s, \pi^*(s)) - Q^*(s, \bar{\pi}(s)) \leq Q^*(s, \bar{\pi}(s)) - \bar{Q}(s, \bar{\pi}(s)) + \bar{Q}(s, \pi^*(s)) - Q^*(s, \pi^*(s))$$

and thus

$$V^*(s) - V_{\bar{\pi}}(s) \leq Q^*(s, \bar{\pi}(s)) - \bar{Q}(s, \bar{\pi}(s)) + \bar{Q}(s, \pi^*(s)) - Q^*(s, \pi^*(s)) \leq 2\epsilon$$

which implies that

$$\|V^* - V_{\bar{\pi}}\|_\infty \leq \frac{2\epsilon}{(1-\gamma)}.$$